

Performance Testing

Date	11 July 2026
Team ID	
Project Name	Rising waters - Flood Prediction using Machine Learning
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman, Browser Developer Tools
Type of Testing	Load Testing & Functional Performance Testing
Target Module / API	Flood Prediction API (Flask)
Test Environment	Render Cloud Deployment
Test Date	11 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single prediction request	1	30	Prediction returned successfully
2	Multiple simultaneous prediction requests	10	60	Stable response without errors
3	Continuous prediction requests	20	120	Consistent performance maintained
4	Invalid input requests	10	30	Validation error handled correctly

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 sec	Pass	Fast prediction
2	Response Time (Max)	< 5 seconds	2.8 sec	Pass	Within acceptable limit
3	Throughput (Req/sec)		15 Req/sec	Pass	Better than expected
4	Error Rate	< 1%	0%	Pass	No request failures
5	CPU Utilization	< 80%	48%	Pass	Stable CPU usage
6	Memory Utilization	< 80%	56%	Pass	Memory usage acceptable

Step 4: Observations & Analysis

Key Findings

The flood prediction system responded quickly under normal and moderate load. All prediction requests were processed successfully with no failures.

Bottlenecks Identified

Minor increase in response time was observed when multiple users submitted requests simultaneously, but performance remained within acceptable limits.

Optimization Steps Taken

Optimized data preprocessing.

Reduced unnecessary computations.

Improved Flask request handling.

Used a trained Random Forest model for faster predictions.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

Flood Prediction

Flood Prediction

Flood Prediction

No Flood